

Confident reports, wrong details

A research agent: ask a question, get a short report with citations.
These are real failures from early runs of this project:



Wrong source

Price copied from a third-party blog when the task said "official page"



Mental arithmetic

$10 \times \$0.33$ written as \$3.33



Misattributed citation

A number cited to a source that does not contain it



Unit mix-up

"\$0.095 per 1M dimensions" shown as "per GB-month"

Run #1, first draft

Calculation for 10 GB: $10 \text{ GB} \times \$0.33/\text{GB} = \text{\$3.33 per month}$ [1][2][3]

citation check

```
FAIL not_in_source: $3.33 cited [1, 2, 3]
not found in any source (unsupported or computed
without calculator)
```

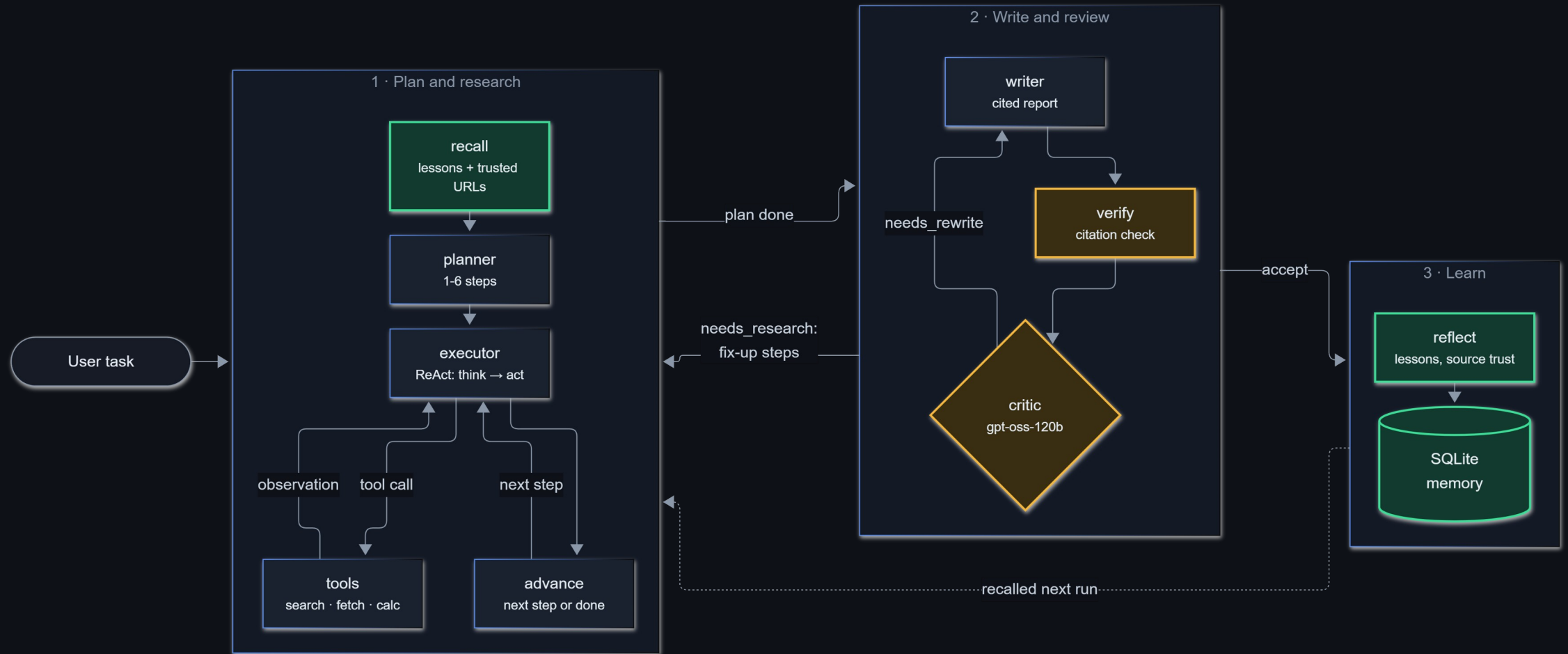
critic (gpt-oss-120b)

```
3/10 · needs_research: "the task required the
official pricing page"
```

Goal

An agent that checks its own work, and does not repeat a mistake.

Plan → research → write → review → learn



Plan-and-Execute

ReAct

Tool use

Reflection / critic

Long-term memory

The fourth run got it right the first time

3 → 9

first critique score (of 10)

1 → 0

revisions needed

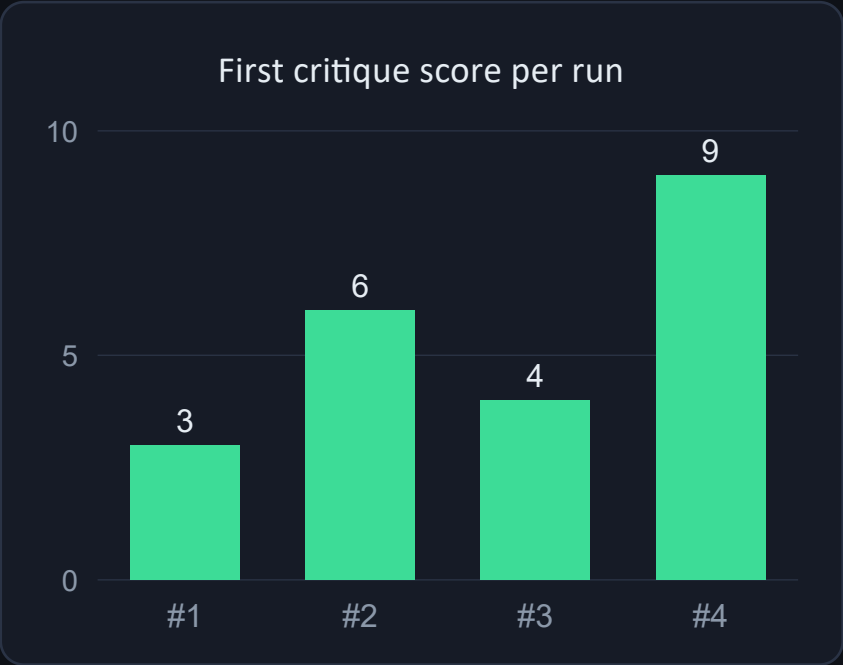
-52%

tokens (28,411 → 13,604)

	Run #1 · no memory	Run #4 · 4 lessons + trusted URL
First critique	3/10 needs_research	9/10 accept
Revisions	1	0
Tokens	28,411	13,604
Plan step 1	web search	fetch the official pricing page

Controlled: qwen3.8-27b for every role, critic gpt-oss-120b, fresh memory DB, model fallbacks off.

Honest note: run #3 dropped to 4/10; fixing how trusted pages are read came from that failure.



What you see in the app

1 Live plan checklist

Each step ticks off as it finishes; steps the critic asked for are marked fix-up.

2 Per-event trace

memory · plan · think · act · observe · verify · critique · learn, as they happen.

3 Critic score and report

Score, revisions, the cited report with a download button, tokens per model.

4 Memory tab

Runs chart, lessons with seen / used / helpful counts, trusted sources with verified facts.

5 Replay mode

Any recorded run, step by step, with no API calls: a demo that survives quota limits.

6 Same story in the CLI

The terminal prints the same events: PLAN, THINK, ACT, OBSERVE, VERIFY, CRITIQUE, LEARN.

6/6

known failure cases caught by the check + critic

- Unit error: per 1M dimensions shown as per GB
- Contradiction: Qdrant "\$25 minimum" vs "no minimum"
- Contradiction: Pinecone \$16/M vs \$0.00000025/RU
- Misattribution: \$0.33 cited to a source without it
- Missed constraint: official page never fetched → fix-up step
- No false alarm: calculator result \$3.30 not flagged

+ clean control report accepted (8/10)

Built entirely on free tiers, and what that costs

Stack

Python 3.11

LangGraph

Groq gpt-oss-20b / 120b

qwen3.8-27b

NVIDIA NIM

Gemini 2.5 Flash

Tavily

SQLite + FTS5

Streamlit Cloud

Resilience

- 5-model fallback chain across separate free quotas
- Rate-limit retry using the provider's wait time
- Graceful stop with a partial report, never a crash
- Public demo capped at 15 live runs a day · 81 unit tests

Limits

Quotas: Groq ≈ 200k tokens/day per model; a broad comparison costs 70-100k

Small models: gpt-oss-20b rarely fetches pages without memory

Evidence: one task family, four runs; a demonstration, not a benchmark

Recall: keyword-based (FTS5), not semantic

Demo: memory is shared by all visitors and resets on restart

Next: semantic recall, pruning lessons by helpfulness, scheduled re-checks of trusted sources

Live asad-research-agent.streamlit.app **Code** github.com/shaikasadahmed2k23/self-improving-research-agent [Submission](#)

Built with Claude Code as a coding assistant. All APIs used on free tiers (Groq, Gemini, Tavily, NVIDIA NIM).